

Invariant Estimator Framework for Autonomous Cars

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I. MATHEMATICAL PRELIMINARIES

A. Notation and Frame Conventions

- ${}^{\mathcal{A}}\mathbf{p}_{\mathcal{B} \rightarrow \mathcal{C}}$ denotes a position vector centered at the origin of reference frame $\mathcal{B}$ extending to the origin of frame $\mathcal{C}$, expressed in frame $\mathcal{A}$. $\mathbf{p}$ denotes an *unknown* position vector, while $\mathbf{r}$ denotes a constant, known value, such as a geometric parameter.
- ${}^{\mathcal{A}}R_{\mathcal{B}} \in SO(3)$ is the rotation matrix relating the orientation of frame $\mathcal{B}$ in frame $\mathcal{A}$.

B. Matrix Lie Groups

- $\mathcal{G} \subset \mathbb{R}^{n \times n}$ denotes the matrix lie group G , with the states $X, Y \in G$ as elements of the group.
- $\mathfrak{g} \subset \mathbb{R}^{n \times n}$ denotes the associated lie algebra for the lie group $\mathcal{G}$.
- $[\cdot]_{\mathcal{G}}^{\wedge} : \mathbb{R}^p \rightarrow \mathfrak{g}$ is the hat operator, a linear isomorphism between the Lie algebra $\mathfrak{g}$ and a p -dimensional vector space, and $[\cdot]_{\mathcal{G}}^{\vee} : \mathfrak{g} \rightarrow \mathbb{R}^p$ is the vee operator, which is the inverse mapping, $\forall \mathbf{a} \in \mathfrak{g}, \mathbf{a} = [\mathbf{a}]_{\mathcal{G}}^{\vee} \in \mathbb{R}^p$, and $\mathbf{a} = [\mathbf{a}]_{\mathcal{G}}^{\wedge}$.
- $\exp_{\mathcal{G}}^{\wedge} : \mathbb{R}^p \rightarrow \mathcal{G}$ is the exponential map from the associated vector space directly to the lie group, such that $\forall \mathbf{a} \in \mathbb{R}^p, \exp_{\mathcal{G}}^{\wedge}(\mathbf{a}) = \exp([\mathbf{a}]_{\mathcal{G}}^{\wedge})$.
- $\log_{\mathcal{G}}^{\vee} : \mathcal{G} \rightarrow \mathbb{R}^p$ is the logarithmic operator that maps the elements of the Lie group $\mathcal{G}$ directly to the vector space, such that $\forall X \in \mathcal{G}, \log_{\mathcal{G}}^{\vee}(X) = \log([X]_{\mathcal{G}}^{\vee})$.
- The group adjoint $\text{Ad}_X : \mathfrak{g} \rightarrow \mathfrak{g}$ is the mapping to the tangent space such that $(\text{Ad}_X \xi)^{\wedge} = X \xi^{\wedge} X^{-1} \in \mathbb{R}^{p \times p}$. The homomorphic property for the combined adjoint Ad_{XY} is defined such that $\text{Ad}_{XY} = \text{Ad}_X \text{Ad}_Y$, and $\text{Ad}_{X^{-1}} = \text{Ad}_X^{-1}$.
- The "small" adjoint $\text{ad}_{\xi} : \mathfrak{g} \rightarrow \mathfrak{g}$ is defined as the derivative of the group adjoint Ad_X at the identity, also known as the Lie bracket, such that $\text{ad}_{\xi}(\eta) = [\xi, \eta] = \xi^{\wedge} \eta^{\wedge} - \eta^{\wedge} \xi^{\wedge}$. Furthermore, the small and group adjoints are related via the exponential, such that $\text{Ad}_{\exp(\mathbf{a}^{\wedge})} = \exp(\text{ad}_{\mathbf{a}})$.

II. CAR STATE

We define the state as in the two-frame group composition, as defined in (Cite Barreau here on this):

$$\chi = \begin{pmatrix} {}^{\mathcal{W}}R_{\mathcal{B}} & {}^{\mathcal{W}}\mathbf{v}_{\mathcal{B}} & {}^{\mathcal{W}}\mathbf{p}_{\mathcal{B}} & \mathbf{b}_g & \mathbf{b}_a & \mathbf{1} \end{pmatrix} \quad (1)$$

s.t. $\chi \in \mathcal{G}_{\text{TF}}(2, 3) \triangleq SO(3) \ltimes (\mathbb{R}^6 \oplus \mathbb{R}^9)$

where $\mathbf{1} \in \mathbb{R}^3$ is an estimate on the lever arm between the GPS and the axle of the car, used for NHC measurement

models CITE which section once you have it. This is an instance of the general two-frame group form $\chi = (R, \mathbf{x}, \mathbf{y})$, where $R \in SO(3)$, $\mathbf{x} \in \mathbb{R}^{3k}$ (world-frame vectors), and $\mathbf{y} \in \mathbb{R}^{3\ell}$ (body-frame vectors). The group's composition properties are defined as the following:

$$\begin{aligned} \chi_1 \chi_2 &= (R_1 R_2, \mathbf{x}_1 + R_1 \mathbf{x}_2, \mathbf{y}_2 + \mathbf{R}_2^{\top} \mathbf{y}_1) \\ \exp(\phi, \mathbf{x}, \mathbf{y}) &= (\Gamma_0(\phi), J_l(\phi) \mathbf{x}, J_r(\phi) \mathbf{y}) \\ \chi^{-1} &= (R^{\top}, -R^{\top} \mathbf{x}, -R \mathbf{y}) \end{aligned} \quad (2)$$

and the exponential $\exp(\phi, \mathbf{x}, \mathbf{y})$ is defined in the appendix. (Add appendix section here.)

As we are defining a primarily **world-centric** estimator, we define the primary error to be in the right-invariant convention:

$$\boldsymbol{\eta} = \hat{\chi} \chi^{-1} \quad (3)$$

Because we have defined the filter state on a matrix lie group, we must define its associated lie algebra. The dimension of the state is 18 (we need only an $\mathbb{R}^3$ perturbation to the orientation as an algebra element). Using the composition properties in (2), we can make use of the group composition properties defined in (2) to decompose (3) into the error for each portion of the state:

$$\begin{aligned} \eta_R &= \hat{R} R^{\top} \\ \eta_{\mathbf{x}} &= \hat{\mathbf{x}} - \eta_R \mathbf{x} \\ \eta_{\mathbf{y}} &= R(\hat{\mathbf{y}} - \mathbf{y}) \end{aligned} \quad (4)$$

This means that the associated lie-algebra coordinates are such that $\boldsymbol{\xi} = (\log \boldsymbol{\eta})^{\vee} \in \mathbb{R}^{18}$.

III. PROCESS MODEL

We define the following continuous time (CT) dynamics:

$$\begin{aligned} {}^{\mathcal{W}}\dot{R}_{\mathcal{B}} &= {}^{\mathcal{W}}R_{\mathcal{B}} {}^{\mathcal{B}}\bar{\boldsymbol{\omega}}^{\wedge} \\ {}^{\mathcal{W}}\dot{\mathbf{v}}_{\mathcal{B}} &= {}^{\mathcal{W}}R_{\mathcal{B}} {}^{\mathcal{B}}\bar{\mathbf{a}} + {}^{\mathcal{W}}\mathbf{g} \\ {}^{\mathcal{W}}\dot{\mathbf{p}}_{\mathcal{B}} &= {}^{\mathcal{W}}\mathbf{v}_{\mathcal{B}} \\ \dot{\mathbf{b}}_g &= \mathbf{w}_{bg}, \mathbf{w}_{bg} \sim \mathcal{N}(0, \Sigma_g) \\ \dot{\mathbf{b}}_a &= \mathbf{w}_{ba}, \mathbf{w}_{ba} \sim \mathcal{N}(0, \Sigma_a) \end{aligned} \quad (5)$$

where ${}^{\mathcal{B}}\bar{\boldsymbol{\omega}}$ and ${}^{\mathcal{B}}\bar{\mathbf{a}}$ are the bias corrected inputs, which are later derived in IV. And we know that these dynamics, when **bias-free**, are *group-affine* - that is, that they yield a log-linear error such that $\boldsymbol{\xi} = A \boldsymbol{\xi}$ (with $\boldsymbol{\eta} = \exp(\boldsymbol{\xi}^{\wedge})$) - this is further shown in the appendix. However, this is generally not the case,

as shown in (CITE Barreau here), that the augmentation of biases onto the InEKF state yields an imperfect InEKF, as $\bar{\omega}$ and $\bar{\mathbf{a}}$ both depend on $\hat{\mathbf{b}}$.

If we assume that the input measurements ${}^{\mathcal{B}}\omega$ and ${}^{\mathcal{B}}\mathbf{a}$ are constant over some small timestep interval $[t, t+\Delta t]$ (i.e., zero-order hold), then we know that the rotation matrix propagation is defined exactly:

$${}^{\mathcal{W}}R_{\mathcal{B}}(t_k + s) = {}^{\mathcal{W}}R_{\mathcal{B},k} \exp(s[\omega]^\wedge) \quad (6)$$

This means that we can substitute this into the discretization of both velocity and position, with $s = u\Delta t$, for some timestep Δt . For instance, with velocity, the integrand yields:

$$\int_0^{\Delta t} \exp(s[\omega]^\wedge) ds = \Delta t \int_0^1 \exp(u[\phi]^\wedge) du = \Delta t \Gamma_1(\phi) \quad (7)$$

And the integral behaves similarly for position. This means that after discretizing, we are left with the following dynamics that obey the $SO(3)$ manifold that live on $SE_2(3)$, with $\phi \triangleq \bar{\omega}\Delta t$, and frames omitted from state components as in (1) for brevity:

$$\begin{aligned} R_{k+1|k} &= R_{k|k} \Gamma_0(\phi) \\ \mathbf{v}_{k+1|k} &= \mathbf{v}_{k|k} + {}^{\mathcal{W}}\mathbf{g}\Delta t + R_{k|k} \Gamma_1(\phi) \mathbf{a} \Delta t \\ \mathbf{p}_{k+1|k} &= \mathbf{p}_{k|k} + \mathbf{v}_{k|k} \Delta t + \frac{1}{2} {}^{\mathcal{W}}\mathbf{g} \Delta t^2 + R_{k|k} \Gamma_2(\phi) \mathbf{a} \Delta t^2 \end{aligned} \quad (8)$$

A. Component-wise Error Derivations

1) *Rotation Error:* Using the right-invariant error convention as defined in (3), we can decompose the rotation error as:

$$\begin{aligned} \dot{\eta}_R &= \dot{R}R^\top + \hat{R}\dot{R}^\top = \hat{R}\bar{\omega}^\wedge R^\top - \hat{R}\omega^\wedge R^\top \\ \dot{\eta}_R &= \hat{R}(\bar{\omega} - \omega)^\wedge R^\top \end{aligned} \quad (9)$$

Using $\bar{\omega} - \omega = -\delta b_g + n_g$, $Ru^\wedge R^\top = (Ru)^\wedge$, and $R\delta b_g = \eta_{b_g}$ gets us

$$\dot{\eta}_R = -\eta_R \eta_{b_g}^\wedge + \eta_R (Rn_g)^\wedge \quad (10)$$

Pattern matching with rotation ate of change gives us $\dot{\eta}_R = -\eta_R \hat{\omega}_b$. We also know that $\omega_b = J_r(\xi_R)\dot{\xi}_R$.

$$\begin{aligned} \dot{\xi}_R &= J_r^{-1}(\xi_R)\omega_b = -J_r^{-1}(\xi_R)\eta_{b_g} \\ &= -J_r^{-1}(\xi_R)J_r(\xi_R)\xi_{b_g} \end{aligned} \quad (11)$$

(12)

2) Body Block Error:

$$\dot{\eta}_y = \dot{R}(\hat{\mathbf{y}} - \mathbf{y}) + R(\dot{\hat{\mathbf{y}}} - \dot{\mathbf{y}}) \quad (13)$$

$\dot{\hat{\mathbf{y}}} = 0$ and $\dot{\mathbf{y}} = W_y$ where W_y is noise.

$$= (R\omega)^\wedge R(\hat{\mathbf{y}} - \mathbf{y}) - RW_y \quad (14)$$

Now we use that

$$R\omega = \eta_R^\top \hat{R}\bar{\omega} + \eta_{b_g} - Rn_g$$

$$\dot{\eta}_y = (\eta_R^\top \hat{R}\bar{\omega} + \eta_{b_g} - Rn_g)^\wedge R(\hat{\mathbf{y}} - \mathbf{y}) - RW_y$$

$$= ((\eta_R^\top \hat{R}\bar{\omega})^\wedge + (\eta_{b_g})^\wedge - (Rn_g)^\wedge) \eta_y - RW_y$$

$$\approx (\hat{R}\bar{\omega})^\wedge \eta_y - RW_y$$

IV. MEASUREMENT MODEL

We formulate the measurement models in both the *right-invariant* and *left-invariant* forms (CITE this), i.e. $\mathbf{Y}_i = X^{-1}\mathbf{d}_i + \mathbf{V}_i$ for right-invariant and $\mathbf{Y}_i = X\mathbf{d}_i + \mathbf{V}_i$ for left-invariant. We define four measurement models, two in the right-invariant form (body frame centric), and two in the left-invariant form.

A. Right-Invariant Measurement Models

First, let us define the error terms.

$$\eta_R^L = R^\top \hat{R}, \quad \eta_{\mathbf{x}}^L = R^\top (\hat{\mathbf{x}} - \mathbf{x}), \quad \eta_y^L = \hat{\mathbf{y}} - \hat{R}^\top R \mathbf{y}$$

1) *Body-Frame Velocity Psuedo Measurement:* We define the body-frame velocity at the IMU origin as ${}^{\mathcal{B}}\mathbf{v} = {}^{\mathcal{W}}R_{\mathcal{B}}^\top {}^{\mathcal{W}}\mathbf{v}_{\mathcal{B}}$. And if we define $\mathbf{r}_w$ as the vector from the IMU origin to the center of the rear axle, by rigid body kinematics we have:

$$\begin{aligned} {}^{\mathcal{B}}\mathbf{v}_{axle} &= {}^{\mathcal{B}}\mathbf{v}_{IMU} + \omega^\wedge \mathbf{r}_w \\ v_{axle} &= R^\top v + \omega^\wedge r_w \end{aligned} \quad (15)$$

We have two cases of ZUPT. Let us tackle the first where the car is moving forward but we are confident that there is no drift.

Let us define a selection matrix $\mathbb{M} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Now let us define a selection matrix for when we know that the car is stopped $\mathbb{M} = I_3$

$$\mathbb{M}v_{axle} = \mathbf{0} + \mathbb{M}n \quad (16)$$

Innovation is trivial $r = -\mathbb{M}(\hat{R}^\top \hat{v} + \bar{\omega}^\wedge r_w)$

Now that we have the equation, we need to put this into Lie group measurement form. As this is body frame, we need to utilize the inverse of the state, as defined in (??). If we define the selection vector $\mathbf{d} = (\mathbf{0}_3, 1, 0)^\top$, then we get our measurement model as:

$$X^{-1}\mathbf{d} = \begin{bmatrix} R^\top & -R^\top \mathbf{v} & -R^\top \mathbf{p} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -R^\top \mathbf{v} \\ 1 \\ 0 \end{bmatrix} \quad (17)$$

2) *Zero Angular Update Rate (ZARU):* This measurement is deliberately excluded from the group, and is simply a method with which to pin $\mathbf{b}_g$ when the car is stationary, as at a zero point we know that $\bar{\omega} = \mathbf{b}_g + \mathbf{n}_g$. Because of this, we can directly formulate the measurement model, along with its linearized Jacobian, as the following:

$$\begin{aligned} \mathbf{z} &= \bar{\omega} - \hat{\mathbf{b}}_g \\ H &= \begin{bmatrix} 0 & 0 & 0 & -\hat{R}_3^\top & 0 \end{bmatrix} \end{aligned} \quad (18)$$

B. Left-Invariant Measurement Models

Left-Invariant measurement models observe sensors attached to the robot that measure in world coordinates.

1) *GPS Position Measurement:* We define the GPS position as the position of the antenna in the ENU(East, North, Up) world frame. Let us define $\mathbf{l}$ as the vector going from the IMU to the GPS-antenna expressed relative to the body-frame.

$$\begin{aligned} z &= p + Rl + n \\ \hat{z} &= \hat{p} + \hat{R}\hat{l} \end{aligned} \quad (19)$$

Where $n \approx \mathcal{N}(0, \text{cov})$. Deriving an expression for true rotation in terms of predicted rotation and error

$$\begin{aligned} \eta_x^L &= {}^{\mathcal{W}}R_B^T(\hat{x} - x) \\ {}^{\mathcal{W}}R_B\eta_x^L &= \hat{x} - x \\ -{}^{\mathcal{W}}R_B\eta_x^L &= x - \hat{x} \\ \text{Now plugging in } x & \\ p - \hat{p} &= -{}^{\mathcal{W}}R_B\eta_p^L \\ \text{Now } p & \\ -\hat{R}\eta_l^L &= \hat{R}\hat{l} - Rl \\ Rl - \hat{R}\hat{l} &= -\hat{R}\hat{l}^L \end{aligned} \quad (20)$$

Now, we derive $z - \hat{z}$

$$\begin{aligned} z - \hat{z} &= p - \hat{p} + Rl - \hat{R}\hat{l} + n \\ z - \hat{z} &= -R\eta_p^L - \hat{R}\eta_l^L + n \\ z - \hat{z} &= -\hat{R}((\eta_R^L)^T \eta_p^L + \eta_l^L) + n \end{aligned} \quad (21)$$

Now let us define the innovation as

$$r = \hat{R}^T(z - \hat{z}) \quad (22)$$

This gives us

$$H_{pos}^L = \begin{bmatrix} 0 & 0 & -I & 0 & 0 & -I \end{bmatrix} \quad (23)$$

Defining our selection vector as $\mathbf{d} = [l^T, 0, 1]^T$:

$$X\mathbf{d} = \begin{bmatrix} R & \mathbf{v} & \mathbf{p} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} l \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} Rl + \mathbf{p} \\ 0 \\ 1 \end{bmatrix} \quad (24)$$

2) *GPS Doppler Velocity Measurement:* We define the GPS velocity as that of the antenna in the ENU world frame. We assume that the gps is rigidly connected to the body. By rigid body kinematics, we have

$$\begin{aligned} {}^{\mathcal{W}}\mathbf{v}_{\text{antenna}} &= {}^{\mathcal{W}}\mathbf{v}_B + {}^{\mathcal{W}}R_B(\omega^\wedge \mathbf{B}\mathbf{l}) + n \\ {}^{\mathcal{W}}\hat{\mathbf{v}}_{\text{antenna}} &= {}^{\mathcal{W}}\hat{\mathbf{v}}_B + {}^{\mathcal{W}}R_B(\bar{\omega}^\wedge \mathbf{B}\hat{\mathbf{l}}) \end{aligned} \quad (25)$$

Where $\bar{\omega}$ is the angular velocity minus the bias $\bar{\omega} = \hat{\omega} - \hat{b}_g$. Now we define some other useful quantities and approximations.

$$\hat{R}^T R = \exp -\xi_R^\wedge = I - \xi_R^\wedge + \frac{1}{2}(\xi_R^\wedge)^2 \dots$$

$$\hat{R}^T R \approx I - \xi_R^\wedge$$

$$\omega = \bar{\omega} + \delta B_g - n_g$$

$$\eta_l^L = \hat{l} - \hat{R}^T Rl$$

$$\eta_R^L \approx I + \xi_R^\wedge$$

$$\eta_l^L \approx \xi_l$$

$$l = \eta_R^L(\hat{l} - \eta_l^L) \approx (I + \eta_R^\wedge)(I - \eta_R) \approx \hat{l} + \xi_R^\wedge \hat{l} - \xi_l$$

$$\delta b_g = \hat{b}_g - b_g \approx \xi_{b_g} - \xi_R^\wedge \hat{b}_g \approx \xi_{b_g}$$

Now let us derive our innovation

$$\begin{aligned} r &= \hat{R}^T(z - \hat{z}) = \\ \hat{R}^T(v - \hat{v}) + \hat{R}^T R(\omega^\wedge l) - \bar{\omega}^\wedge \hat{l} + \hat{R}^T n &= \\ \text{plugging in our approximations gives us} & \\ (I - \xi_R^\wedge)(\bar{\omega} + \xi_{b_g} - n_g)^\wedge (\hat{l} + \xi_R^\wedge \hat{l} - \xi_l) - \bar{\omega}^\wedge \hat{l} &= \\ \text{Now multiplying out the parenthesis and} & \\ \text{removing terms with two small values gives us} & \\ \bar{\omega}^\wedge \hat{l} - \xi_R^\wedge \bar{\omega}^\wedge \hat{l} + \xi_{b_g}^\wedge \hat{l} - n_g^\wedge \hat{l} + \bar{\omega} \xi_R^\wedge \hat{l} - \bar{\omega} \xi_l - \wedge \hat{l} &= \\ \xi_R^\wedge \bar{\omega}^\wedge \hat{l} + \xi_{b_g}^\wedge \hat{l} - n_g^\wedge \hat{l} + \bar{\omega} \xi_R^\wedge \hat{l} - \bar{\omega} \xi_l &= \\ -(\xi_R^\wedge \bar{\omega}^\wedge - \bar{\omega}^\wedge \xi_R^\wedge) \hat{l} - \hat{l} \xi_{b_g} - \bar{\omega} \xi_l - n_g^\wedge \hat{l} &= \\ -\hat{l}^\wedge \bar{\omega}^\wedge \xi_R - \hat{l}^\wedge \xi_{b_g} - \bar{\omega} \xi_l - n_g^\wedge \hat{l} \end{aligned} \quad (26)$$

This gives us

$$H_{dopVel}^L = \begin{bmatrix} -\hat{l}^\wedge \bar{\omega}^\wedge & -I & 0 & -\hat{l}^\wedge & 0 & -\bar{\omega} \end{bmatrix} \quad (27)$$

We define our selection vector as $\mathbf{d} = (0_3, 1, 0)$

$$X\mathbf{d} = \begin{bmatrix} R & \mathbf{v} & \mathbf{p} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} v \\ 1 \\ 0 \end{bmatrix} \quad (28)$$

C. Update

Our filter propagates variance using right-invariant error $\eta = \hat{\chi}\chi^T$. Our pseudo measurements are right-invariant and can be applied directly to the error. The GNSS measurements however are left invariant and have constant (or close to constant for biases) jacobians only in left-invariant coordinates.